Discussion

Tuesday, October 6, 2020 2:47 PM

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OH: Tue 10-11 am (right after discussion)
Friday (8-9 am) (tentative)

Lecture notes and recordings: Upload CCLE
↳ unless significant drop in attendance.

What is probability? Is a method to quantify the possibility of occurence of events

We have a class of students.

	Boys	Girls
Blue	5	10
Green.	15	20

Assume the teacher choses one student (S) uniformly random (meaning each student have chance to be chosen) from this class.

What is the prob. that S has green eyes? $P(B) = ?$

What is the probability that S is a boy? $P(A) = ?$

↳ $A = \{S \text{ is boy}\}$ ↳ $B = \{S \text{ has green eyes}\}$

$\hookrightarrow A = \{S \text{ is boy}\}$ $\hookrightarrow B = \{S \text{ has green eyes}\}$

	Boys	Girls
Blue	5	10
Green	15	20

$P(A) = \frac{20}{50} = \frac{2}{5}$

$P(B) = \frac{35}{50} = \frac{7}{10}$

$P(A \cap B) = P(A \text{ and } B \text{ occurs at the same time})$

$= (S \text{ is boy } \& \ S \text{ has green eyes})$

$= \frac{15}{50}$

$P(A \cup B) = P(A \text{ or } B \text{ occurs})$

$= \frac{5+15+20}{50} = \frac{40}{50} = \frac{4}{5}$

$P(A') = P(A^c) = P(S \text{ is not boy}) = \frac{30}{50} = \frac{3}{5}$

complement of set A

Ex: 1.1.7: Given that $P(A \cup B) = 0.76$, $P(A \cup B') = 0.87$,

what is $\boxed{P(A)?}$

A) .24 B) .37 C) .50 D) .63 E) .77

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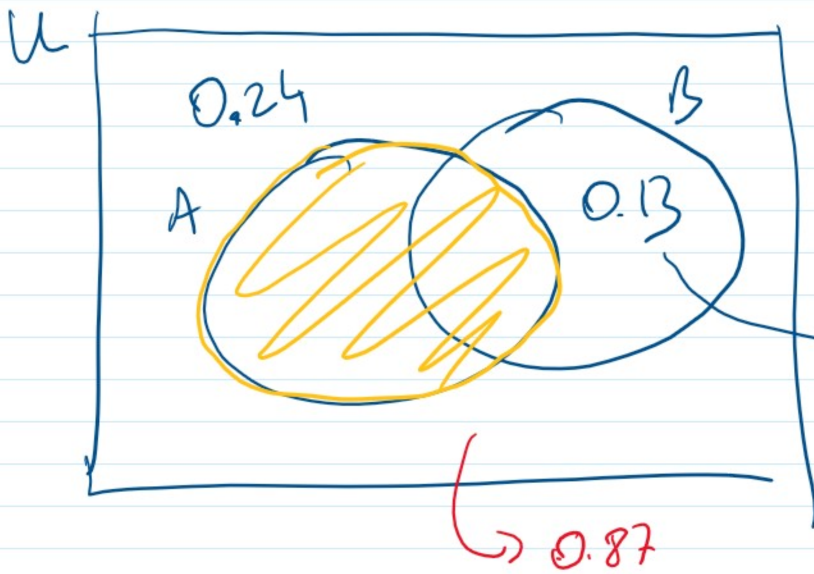


$P((A\cup B)') = 1 - 0.76$
$= 0.24$

$P((A\cup B')') = P(B\setminus A)$
$= 1 - 0.87$
$= 0.13$

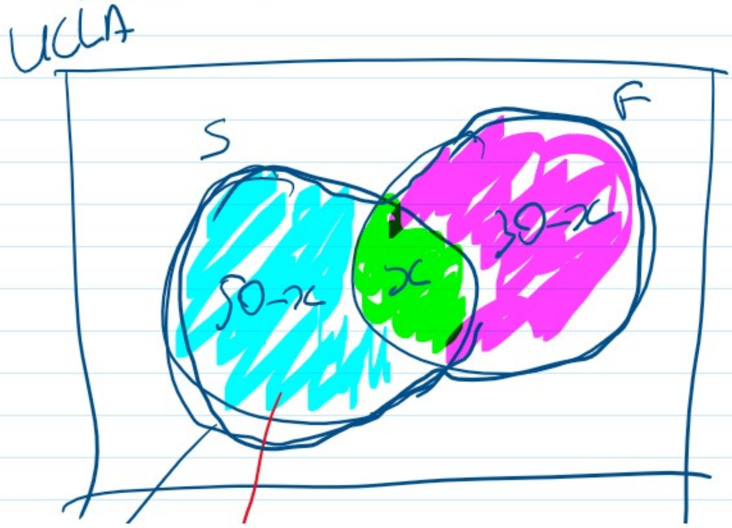
$P(A') = 0.24 + 0.13 = 0.37$

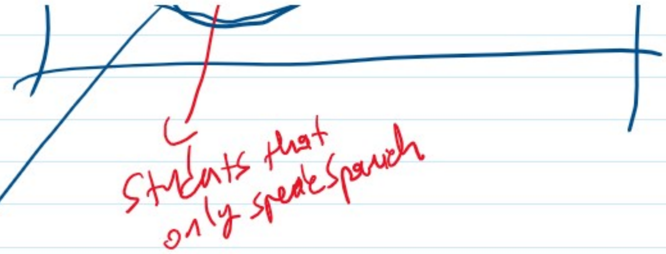
$1 - P(A) = 0.37 \Rightarrow P(A) = 1 - 0.37 = \boxed{0.63}$

Ex: In UCLA, 70 percent of students can speak either French or Spanish. Half of UCLA students can speak Spanish and 30% can speak French. Find the percentage of students that can speak both.

A) 5 $\boxed{\text{B) } 10}$ C) 20 D) 40 E) 30

Sol





Total region must add up to 70%

$$50-x+x+30-x=70$$

$$80-x=70$$

$$\boxed{x=10}$$

Thm: Principle of Inclusion & Exclusion (PIE)

- Let A & B be 2 arbitrary sets. Then

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$

Proof:

Yellow + Green.

$$P(A)=P(A\setminus B)+P(A\cap B)$$

$$P(B)=P(B\setminus A)+P(A\cap B)$$

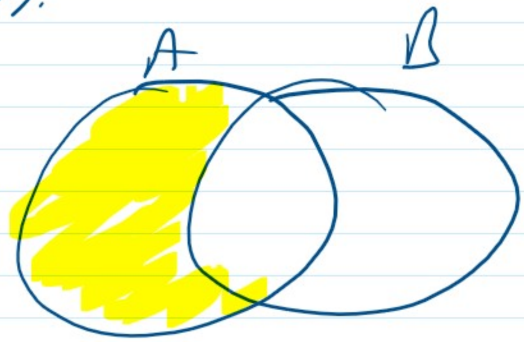
Purple + Green

$P(A)+P(B)$ double counts $P(A\cap B)$

$$P(A)+P(B)=Y+G+G+P=P(A\cup B)+P(A\cap B)$$

$Y+G+P \to P(A\cup B)$, $\quad G \to P(A\cap B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$.



$A \setminus B = A - B$,

represents all elements which are in A but not in B

(PIE for 3 sets) A, B, C are 3 arbitrary sets, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Ex: In UCLA, 70 percent of students can speak either French or Spanish. Half of UCLA students can speak Spanish, and 30% can speak French. Find the percentage of students that can speak both.

A) 5 $\quad \boxed{\text{B) 10}}$ $\quad$ C) 20 $\quad$ D) 40 $\quad$ E) 30

Second solution: $S = \{\text{students that can speak spanish}\}$

$F = \{\text{students that can speak french}\}$

$P(S \cup F) = 0.7$

$P(S) = 0.5 \qquad P(F) = 0.3 \qquad P(S \cap F)$

$P(S \cup F) = P(S) + P(F) - P(S \cap F)$.

$0.7 = 0.5 + 0.3 - \underline{P(S \cap F)}$

$$0.2 = 0.5 + 0.3 - \underbrace{P(S \cap F)}_{0.10.}$$

Ex (1.1.1) Of group of patients having injuries, 28% visit both physical therapist (PT) and a chiropractor (Ch) and 8% visit neither. Say that prob. of visiting PT exceeds the prob. of visiting Ch. by 16%, what is the prob. of a randomly selected patient from this group visit PT?

A) 0.24 B) 0.38 C) 52 D) 68 E) 72